

MATH 301: Homework 1

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Question 1: From Blackboard

Proposition: Let S and T be two sets. If both S and T are countable, then $S \cup T$ is countable.

Note: If S and T are said to be infinite and countable; then they are said to be *denumerable*

Proof. Suppose S and T are countable, and there exists $A, B \subseteq \mathbb{N}$, by definition there exists a bijective function:

$$f : A \rightarrow S \text{ and } g : B \rightarrow T$$

To show $S \cup T$ is countable, we must construct a bijection $h : C \rightarrow S \cup T, C \subseteq \mathbb{N}$

Define $h : C \rightarrow S \cup T$:

$$h(n) = \begin{cases} f\left(\frac{n}{2}\right) & \text{if } n \text{ is even} \\ g\left(\frac{n+1}{2}\right) & \text{if } n \text{ is odd} \end{cases}$$

Note S, T must be disjoint; else $\exists n_1, n_2 \in C, n_1 \neq n_2 \ni h(n_1) = f(n_1) = h(n_2) = g(n_2)$

Thus, for the sake of the proof, presume S, T to be disjoint sets s.t. $S \cap T = \emptyset$

To prove the proposition we must show that h is *surjective* and *injective*.

Surjectivity

Proof. **Goal:** We want to show that $\forall y \in S \cup T, \exists n \in C \ni h(n) = y$

Case 1 ($y \in S$):

Let $y \in S$. Since f is surjective, $\exists k \in C$ s.t. $f(k) = y$.

Let $n = 2k$. By definition n is even. Observe:

$$h(n) = f\left(\frac{2k}{2}\right) = f(k) = y$$

Case 2 ($y \in T$):

Let $y \in T$. Since g is surjective, $\exists k \in C \ni g(k) = y$

Let $n = 2k - 1$. Then n is odd. Observe:

$$h(n) = g\left(\frac{(2k-1)+1}{2}\right) = g\left(\frac{2k}{2}\right) = g(k) = y$$

As every element $y \in S \cup T \ni n \in C \ni h(n) = y \implies h : C \rightarrow S \cup T$ is surjective. □

Injectivity

Proof. **Goal:** We want to show that for arbitrary elements $n_1, n_2 \in C, h(n_1) = h(n_2) \implies n_1 = n_2$.
Suppose $h(n_1) = h(n_2) = y$.

Case 1 (Same Parity):

Suppose n_1, n_2 be both even. Then the following statements hold.

$$h(n_1) = h(n_2) \implies f\left(\frac{n_1}{2}\right) = f\left(\frac{n_2}{2}\right).$$

$$\text{Since } f \text{ is injective: } \frac{n_1}{2} = \frac{n_2}{2} \implies n_1 = n_2$$

Suppose n_1, n_2 be both odd. Then the following statements hold.

$$h(n_1) = h(n_2) \implies g\left(\frac{n_1+1}{2}\right) = g\left(\frac{n_2+1}{2}\right)$$

$$\text{Since } g \text{ is injective: } \frac{n_1+1}{2} = \frac{n_2+1}{2} \implies n_1 = n_2$$

Case 2 (Mixed Parity):

WLOG suppose n_1 is even and n_2 is odd.

$$h(n_1) \in \text{Img}(f) = S$$

$$h(n_2) \in \text{Img}(g) = T$$

$$\text{Since } S \cap T = \emptyset \implies h(n_1) \neq h(n_2) \equiv f(n_1) \neq g(n_2)$$

As $n_1, n_2 \in C$, $h(n_1) = h(n_2) \implies n_1 = n_2$. The function $h(n) : C \rightarrow S \cup T$ is injective □

Conclusion

Since h is both surjective and injective, it is bijective.

As there exists a bijective function $h : C \rightarrow S \cup T \implies S \cup T$ is countable. □

Question 2: Sec 1.3 (q1)

Question: Prove that a nonempty set T_1 is finite iff $\exists$ a bijection from T_1 onto a finite set T_2 .

Case 1 $\rightarrow$:

Note: To prove this by "Pigeon Hole Principle" we must suppose $|T_1| = |T_2|$ else there does not exist a way to construct a bijection.

Proof. (**Show:** If T_1 is finite. Then $\exists$ a $f : T_1 \rightarrow T_2$, where T_2 is finite and f is bijective)

Suppose T_1 is finite. Consider a function $f : T_1 \rightarrow T_2$ defined as:

$$f := \{(a_i, b_j) | a_i \in T_1 \text{ and } b_j \in T_2\}$$

Let a_i be an arbitrary element defined to be (where all elements are distinct):

$$a_i = a_1, a_2, a_3, \dots, a_{|T_1|},$$

Let b_i be an arbitrary element defined to be (where all elements are distinct):

$$b_i = b_1, b_2, b_3, \dots, b_{|T_2|},$$

Clearly:

$$\forall b \in T_2, \exists a \in T_1 \ni f(a) = b$$

Thus f is surjective.

Clearly:

$$f(x) = f(y) \implies x = y$$

Thus f is injective

Conclusion: As f is both injective and surjective, given T_1 is finite then clearly there exists a bijection from T_1 onto T_2 . □

Case 2 $\leftarrow$:

Proof. (**Show:** If $\exists$ a $f : T_1 \rightarrow T_2$, where T_2 is finite and f is bijective. Then T_1 is finite.)

Suppose $\exists$ a $f : T_1 \rightarrow T_2$, where T_2 is finite and f is bijective.

For the sake of contradiction suppose T_1 is infinite.

As f is bijective we can construct an arbitrary mapping as:

$$f := \{(a_i, b_j) | a_i \in T_1 \text{ and } b_j \in T_2\}$$

Let a_i be an arbitrary element defined to be (where all elements are distinct):

$$a_i = a_1, a_2, a_3, \dots, a_{|T_1|},$$

Let b_i be an arbitrary element defined to be (where all elements are distinct):

$$b_i = b_1, b_2, b_3, \dots, b_{|T_2|},$$

However as T_1 is infinite and T_2 is finite then clearly $|T_2| < |T_1|$.

Thus for the function $f : T_1 \rightarrow T_2$ to be a function the following must be true (by "Pigeon Hole Principle"):

$$f(x) = f(y) \implies x \neq y$$

This is a contradiction, thus we can say: if $\exists$ a $f : T_1 \rightarrow T_2$, where T_2 is finite and f is bijective. Then T_1 is finite. $\square$

Question 3: Sec 1.3 (q4)

Proof. **Question:** Exhibit a bijection between $\mathbb{N}$ and the set of all odd integers greater than 13.

Setup

We want to find a set of *all odd integers greater than 13*. Let's define this as S (where $i \in \mathbb{N}$):

$$S = \{15, 17, 19, \dots\} = \{2z + 13 | z \in \mathbb{N}\}$$

(Note: all elements defined are distinct)

Also let's define elements of $\mathbb{N}$ as follows, $j \in \mathbb{N}$:

$$n_1 = 1$$

$$n_2 = 2$$

$$n_3 = 3$$

$$\dots$$

$$n_j = j$$

(Note: all elements defined are distinct)

Let's construct a function that maps $\mathbb{N}$ to S :

$$f : \mathbb{N} \rightarrow S = \{(s_i, n_j) | s_i \in S \wedge n_j \in \mathbb{N}\}$$

Observe this is modeling the function as:

$$f(n) = 13 + 2n$$

Surjectivity

Proof. Goal: Show that $\forall s \in S, \exists n \in \mathbb{N} \ni f(n) = s$

Let $s \in S, n \in \mathbb{N}$:

$$s_n = 13 + 2n = f(n)$$

By algebra:

$$s_n = 13 + 2n$$

$$2n = s_n - 13$$

Recall $\forall s \in S$, s must be an odd integer greater than 13. Thus there must be some $k \in \mathbb{N}$ s.t.

$$s = 2k + 1$$

By substitution:

$$2n = s_n - 13$$

$$2n = (2k + 1) - 13$$

$$2n = 2k + (1 - 13)$$

$$2n = 2k - 12$$

$$n = k - 6$$

Now we must ask if n as defined to be as $n = k - 6$ escapes the set $\mathbb{N}$ as n is defined to be $n \in \mathbb{N}$. As s is defined to be *greater than 13*. We know there exists a greatest lower bound 15 that satisfies the criterion. By substitution:

$$s > 13$$

$$s = 2k + 1$$

$$2k + 1 > 13$$

$$2k + 1 - 1 > 13 - 1$$

$$2k > 12$$

$$k > 6$$

As $k > 6$ then this means the $n = k - 6$ must always be greater than zero. As all operations retain closure properties of integers, we can expect the value to be an integer. By def all integers greater than zero must belong to $\mathbb{N}$. Thus we can say $n \in \mathbb{N}$. Thus the properties of surjectivity hold.

Backsolve:

$$f(k - 6) = 13 + 2(k - 6)$$

$$f(k - 6) = 13 + 2k - 12$$

$$f(k - 6) = 13 - 12 + 2k$$

$$f(k - 6) = 1 + 2k = s$$

We have shown that $\forall s \in S, \exists n \in \mathbb{N} \ni f(n) = s$

□

Injectivity

Proof. Goal: Show that given $a, b \in \mathbb{N}$ and $f(a) = f(b) \implies a = b$

Let $a, b \in \mathbb{N}$. Suppose:

$$f(a) = f(b)$$

By substitution:

$$\begin{aligned} 2a + 1 &= 2b + 1 \\ 2a + 1 - 1 &= 2b + 1 - 1 \\ 2a &= 2b \\ a &= b \end{aligned}$$

Thus we have shown that given $a, b \in \mathbb{N}$ and $f(a) = f(b) \implies a = b$

□

Conclusion

As f is both injective and surjective we have shown there exists a bijection between $\mathbb{N}$ and the set of all odd integers greater than 13. □

Question 4: Sec 1.3 (q6)

Question: Exhibit a bijection between $\mathbb{N}$ and a proper subset of itself.

Let's construct a set E , a set of all even natural numbers, and let $k \in \mathbb{N}$:

$$E := \{n | n = 2k\}$$

Clearly $1 \in \mathbb{N}$ and $1 \notin E$ as 1 is not even. Therefore $E \subset \mathbb{N}$.

To prove a bijection exists, we must construct a function $f : \mathbb{N} \rightarrow E$ that is surjective and injective. Let's construct the function as follows, let $n \in \mathbb{N}, v \in E$:

$$f(n) = 2n = v$$

Surjectivity

Proof. Goal: Show that $\forall s \in S, \exists n \in \mathbb{N} \ni f(n) = s$

Let $v \in E, n \in \mathbb{N}$. We can say the following:

$$v = 2n \implies \frac{v}{2} = n$$

Recall that $v \in E$, which constricts the elements of E to be even, as defined. Let $k \in \mathbb{N}$ and $v = 2k$. Thus we can rearrange the equation as follows:

$$\frac{v}{2} = n \implies \frac{2k}{2} = n \implies k = n$$

Now notice we have not violated any closure properties thus the solution remains valid.

Backsolve: to confirm validity.

$$f(k) = 2k = v$$

Recall: we had defined $v = 2k$, so this is the expected answer.

Conclusion:

Thus, we have show that $\forall s \in S, \exists n \in \mathbb{N} \ni f(n) = s$
Therefore f is surjective.

□

Injectivity

Proof. Goal: Show that given $a, b \in \mathbb{N}$ and $f(a) = f(b) \implies a = b$
Let $a, b \in \mathbb{N}$ and $f(a) = f(b) \implies a = b$. Observe:

$$\begin{aligned} f(a) &= f(b) \\ 2a &= 2b \\ \frac{2a}{2} &= \frac{2b}{2} \\ a &= b \end{aligned}$$

Thus we have shown that given $a, b \in \mathbb{N}$ and $f(a) = f(b) \implies a = b$
Therefore f is injective.

□

Conclusion

As f is both injective and surjective we have shown there exists a bijection between $\mathbb{N}$ and the set of all odd integers greater than 13.

Question 5: Sec 1.3 (q7)

Question: Prove that a set T_1 is denumerable iff there is a bijection from T_1 onto a denumerable set T_2

Case 1 $\rightarrow$:

Prove: Given a denumerable set T_1 we can construct a bijection onto a denumerable set T_2 .

Proof. Suppose T_1 is denumerable. This implies a bijective function f exists:

$$f : \mathbb{N} \rightarrow T_1$$

Consider the existance of another denumerable set T_2 . Likewise a bijective fuction g must also exist:

$$g : \mathbb{N} \rightarrow T_2$$

As these functions are bijective, this implies the existance of an inverse function for each.

$$f^{-1} : T_1 \rightarrow \mathbb{N} \text{ and } g^{-1} : T_2 \rightarrow \mathbb{N}$$

Goal: we want to show there exists a function $h : T_1 \rightarrow T_2$ that is bijective.

Approach: consider the change in space

$$T_1 \rightarrow \mathbb{N} \rightarrow T_2$$

Consider the composition function:

$$g \circ f^{-1} : T_1 \rightarrow T_2$$

From definition above we know $\text{Img}(f^{-1}) = \mathbb{N}$, via the implication of bijection.

We also know that $g : \mathbb{N} \rightarrow T_2$ is bijective.

Thus we can say the following is a valid bijective function:

$$g : \text{Img}(f^{-1}) \rightarrow T_2 \equiv g : \mathbb{N} \rightarrow T_2$$

Conclusion: Thus we have shown given a denumerable set T_1 we can construct a bijection towards another denumerable set T_2 . □

Case 2 $\leftarrow$:

Prove: Given a denumerable set T_2 , if a bijection exists from T_1 onto T_2 then T_1 must be denumerable.

Proof. Suppose T_2 is a denumerable set. Thus, there must exist a bijection from T_2 and $\mathbb{N}$. Thus there must exist some functions:

$$f : \mathbb{N} \rightarrow T_2 \text{ and } f^{-1} : T_2 \rightarrow \mathbb{N}$$

Suppose there exists a bijection from T_1 onto T_2 . We can define the following functions:

$$g : T_1 \rightarrow T_2 \text{ and } g^{-1} : T_2 \rightarrow T_1$$

Approach: consider the change in space

$$T_1 \rightarrow T_2 \rightarrow \mathbb{N}$$

We can construct the composition:

$$f^{-1} \circ g : T_1 \rightarrow \mathbb{N}$$

Consider the image set:

$$\text{Img}(g) = T_2$$

By substitution:

$$f^{-1}(\text{Img}(g)) = \mathbb{N}$$

As these functions are bijective, we can consider their inverses.

$$f(\mathbb{N}) = T_2 \implies g(T_2) = T_1 \implies g(f(\mathbb{N})) = T_1$$

This means there exists a bijective function from $\mathbb{N} \rightarrow T_1$. Therefore, T_1 must be denumerable.

Conclusion: We have shown given a denumerable set T_2 , if a bijection exists from T_1 onto T_2 then T_1 must be denumerable. □

Question 6: Sec 1.3 (q8)

Question: Give an example of a countable collection of finite sets whose union is not finite. Consider the sets defined to be:

$$A_n = \{n\}, \forall n \in \mathbb{N}$$

These sets are finite sets defined as:

$$A_1 = \{1\}$$

$$A_2 = \{2\}$$

$$A_3 = \{3\}$$

...

$$A_n = \{n\}$$

Observe each set $A_1, A_2, A_3, \dots, A_n$ are finite (singleton) sets.

Though the union of all sets, by how we defined $A_1, A_2, A_3, \dots, A_n$, would be result in the equivalent set $\mathbb{N}$ which is not finite.